Use of biological resources

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1

Paper 1 & 2 | Food production | [1]

Maya investigates how the type of sugar affects anaerobic respiration in yeast. She mixes yeast with a sugar solution in a test tube, adds a layer of oil on top and counts the bubbles of gas released in 5 minutes.

Which control experiment would show that the gas is produced by the living yeast?

A. the same set-up with twice the mass of yeast

B. the same set-up without the layer of oil

C. the same set-up with yeast that has been boiled and then cooled

D. the same set-up kept at a higher temperature

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2 · Part 1

Paper 1 & 2 | Food production | [1]

In an industrial fermenter, cold water is pumped through a water jacket that surrounds the vessel. The water does not mix with the contents of the fermenter.

Why is the water jacket needed?

A. to add water to the culture so that the micro-organisms can grow

B. to remove heat released by respiration of the micro-organisms, so the temperature stays near the optimum

C. to cool the culture enough to kill any unwanted micro-organisms

D. to keep the culture cold so that the micro-organisms stop dividing

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3 · Part 1 Paper 2 only Food production [1]

Fish farmers try to control interspecific predation.

Which method controls interspecific predation?

A keeping young fish and adult fish of the same species in separate tanks

B feeding the fish small amounts of food several times a day

C adding antibiotics to the water

D covering outdoor ponds with nets to keep out fish-eating birds

4 · Part 1 Paper 1 & 2 Selective breeding [1]

Farmers have selectively bred one breed of dairy cattle for high milk yield for many generations.

Why could this cause a problem for the farmers?

A The breed has little genetic variation, so a new disease could affect most of the animals.

B The cattle have become transgenic, so their milk cannot be sold.

C The cattle are all clones, so they cannot reproduce sexually.

D The high milk yield will not be passed on to the next generation.

5 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [1]

Which organism is transgenic?

A a strawberry plant grown from a runner of its parent plant

B a sheep produced by cloning a cell from another sheep

C a bacterium that has had a human gene inserted into its plasmid

D a wheat plant produced by crossing two wheat varieties for many generations

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6 · Part 1

Paper 1 & 2 | Genetic modification (genetic engineering) | [1]

Why can viruses be used as vectors in genetic modification?

A. They insert their genetic material into the cells that they infect.

B. They contain plasmids, which are found only in viruses.

C. They make restriction enzymes that cut the DNA of the host cell.

D. They reproduce by cell division inside the nucleus of the host cell.

7 · Part 1

Paper 2 only | Cloning | [1]

Plants produced by micropropagation from one parent plant are genetically identical.

What is the reason for this?

A. The new cells are produced by meiosis.

B. The explants are grown in a sterile medium, so no mutations can happen.

C. The explants are fertilised by pollen from the same parent plant.

D. The new cells are produced by mitosis from cells of one parent plant.

8 · Part 1

Paper 2 only | Cloning | [1]

In micropropagation, explants are grown *in vitro*.

What does *in vitro* mean?

A. in soil inside a glasshouse

B. in a sterile container in a laboratory, outside a living organism

C. inside the tissues of the parent plant

D. in a field covered by a polythene tunnel

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9 · Part 1

Paper 1 & 2 | Food production | [8]

Tomato plants were grown in four glasshouses. The light intensity was the same in all four glasshouses. The table shows the carbon dioxide concentration, the temperature and the yield of tomatoes in each glasshouse.

Glasshouse	Carbon dioxide concentration / %	Temperature / °C	Yield / kg per plant
1	0.04	15	4.0
2	0.10	15	5.2
3	0.04	25	5.0
4	0.10	25	7.0

(a) Describe two ways, other than increasing the carbon dioxide concentration and the temperature, in which growing crops in a glasshouse can increase their yield. [2]

(b) Calculate the percentage increase in yield from glasshouse 1 to glasshouse 4.

Use the formula: percentage increase = (new value − original value) ÷ original value × 100 [2]

(c) Explain why increasing the carbon dioxide concentration and increasing the temperature both increased the yield of tomatoes. [3]

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(d) Suggest one reason why the grower does not increase the temperature in the glasshouse to 45 °C. [1]

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10 · Part 1

Paper 1 & 2 Food production [7]

Wheat was grown in fields given different masses of a fertiliser containing nitrate ions. All other conditions were kept the same. The table shows the results.

Mass of fertiliser added / kg per hectare	Yield of wheat / tonnes per hectare
0	3.0
50	4.6
100	5.8
150	6.4
200	6.5
250	6.5

(a) Describe the effect of the mass of fertiliser added on the yield of wheat. [2]

(b) Farmer Ortiz has a wheat field with an area of 12 hectares.

Calculate how many more tonnes of wheat this field would produce with 150 kg per hectare of fertiliser than with no fertiliser. [2]

(c) Explain how the nitrate ions in the fertiliser increase the yield of wheat. [2]

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(d) Suggest why Farmer Ortiz decides to add 150 kg per hectare of fertiliser rather than 250 kg per hectare. [1]

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11 · Part 1

Paper 1 & 2 | Food production | [7]

Whitefly are small insects that feed on the sap of tomato plants grown in glasshouses.

Chen tested two ways of controlling whitefly in two identical glasshouses, P and Q.

- Glasshouse P was sprayed once with a pesticide in week 0.
- In glasshouse Q, a small wasp was released in week 0. The wasp lays its eggs inside young whitefly, and the wasp larvae kill the whitefly. The wasp does not feed on tomato plants.

The table shows the mean number of whitefly per leaf in each glasshouse.

Week	**0**	**2**	**4**	**6**	**8**	**10**
Glasshouse P (pesticide)	40	4	10	22	34	41
Glasshouse Q (wasp)	40	31	18	9	6	5

(a) Give one reason why growers need to control pests on crop plants. [1]

(b) Use the data to describe how the effect of the pesticide on the number of whitefly differs from the effect of the wasp. [2]

(c) Discuss the advantages and disadvantages of using biological control rather than pesticides to control pests on crop plants. [4]

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12 · Part 1 Paper 1 & 2 Food production [8]

Bread dough is made by mixing flour, water, sugar and yeast. The dough is left in a warm place for one hour and then baked in an oven at 220 °C. In the dough, the yeast respires anaerobically.

(a) Give the word equation for anaerobic respiration in yeast. [2]

(b) Explain how anaerobic respiration by the yeast makes the dough rise. [2]

(c) Explain why the dough is left in a warm place rather than in a cold place. [2]

(d) Explain why the dough stops rising once it has been in the oven for a few minutes. [2]

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13 · Part 1

Paper 1 & 2 | Food production | [9]

Ravi investigates the effect of temperature on anaerobic respiration in yeast.

- He puts 10 cm^3 of yeast suspension and 10 cm^3 of glucose solution into a boiling tube.
- He adds a layer of oil on top of the mixture.
- He connects the boiling tube to a gas syringe with a delivery tube.
- He places the boiling tube in a water bath at the test temperature for 5 minutes.
- He then measures the volume of gas collected in the gas syringe in the next 10 minutes.

He does three trials at each of four temperatures. The table shows his results.

Temperature / °C	**Trial 1 volume / cm^3**	**Trial 2 volume / cm^3**	**Trial 3 volume / cm^3**	**Mean volume / cm^3**
20	4.2	4.6	4.4	4.4
30	9.8	10.4	10.1	
40	16.0	15.4	15.8	15.7
50	6.1	5.5	5.8	5.8

(a) Explain why Ravi adds a layer of oil on top of the mixture. [2]

(b) Calculate the mean volume of gas collected at 30 °C. [1]

(c) Calculate the rate of gas production at 40 °C in cm^3 per minute.

Use the formula: rate = mean volume of gas ÷ time [2]

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(d) State the dependent variable in this investigation. [1]

(e) State one other variable that Ravi should keep constant. Do not give temperature, time or volume of solution. [1]

(f) Explain why the mean volume of gas is lower at 50 °C than at 40 °C. [2]

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14 · Part 1

Paper 1 & 2

Food production

[7]

The table shows the steps used by a dairy to make yoghurt. Milk contains a sugar called lactose.

Step	What is done
1	Milk is heated to 85 °C for 15 minutes.
2	The milk is cooled to 42 °C.
3	*Lactobacillus* bacteria are added to the milk.
4	The milk is kept at 42 °C for 6 hours.
5	The yoghurt is cooled to 5 °C and stored.

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(a) Explain why the milk is heated to 85 °C in step 1. [2]

(b) Explain why the milk is cooled to 42 °C before the bacteria are added. [2]

(c) Describe what the *Lactobacillus* bacteria do during step 4 that turns the milk into yoghurt. [3]

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15 · Part 1

Paper 1 & 2 | Food production | [8]

Food company Mycofoods grows a fungus in an industrial fermenter to make a protein-rich food.

The fermenter is a large steel vessel. Before use, it is cleaned and sterilised with steam. Sterile air is bubbled in through a pipe at the bottom. Paddles turned by a motor stir the contents. Probes measure the temperature and the pH. Nutrients, including glucose and ammonium salts (a source of nitrogen), are added through an inlet pipe.

(a) Explain why sterile air is bubbled into the fermenter. [2]

(b) Explain why the contents of the fermenter are stirred by the paddles. [2]

(c) Explain why the fungus is supplied with glucose and with a source of nitrogen. [2]

(d) When the probe shows that the pH has changed, acid or alkali is added to the fermenter. Explain why the pH must be kept at the optimum. [2]

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16 · Part 1

Paper 2 only

Food production

[9]

Tilapia are fish that are farmed in tanks as a source of protein. A fish farmer kept tilapia in tanks at different stocking densities (number of fish per m^3 of water) for 6 months. All the fish were given the same food.

The table shows the results.

Stocking density / fish per m^3	Mean increase in mass per fish / g	Fish with disease / %
10	820	2
20	760	5
40	610	14
80	430	31

(a) Use the data to describe the effects of increasing the stocking density. [2]

(b) Suggest two reasons why the percentage of fish with disease is higher at high stocking densities. [2]

(c) Explain why fish farmers remove waste products and uneaten food from the water in the tanks. [2]

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(d) Describe how selective breeding could be used to produce tilapia that grow faster. [3]

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17 · Part 1

Paper 1 & 2 | Selective breeding | [6]

Rust is a disease of wheat caused by a fungus. A plant breeder has two varieties of wheat.

- Variety P has a high yield but is not resistant to rust.
- Variety Q has a low yield but is resistant to rust.

(a) Describe how the plant breeder could use selective breeding to develop a new variety of wheat that has a high yield and is resistant to rust. [3]

(b) Explain why the breeder needs to repeat the process for many generations. [2]

(c) Suggest why the breeder grows the young wheat plants in a field where the rust fungus is present. [1]

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18 · Part 1

Paper 1 & 2 Selective breeding [5]

Poultry farmer Amara used selective breeding to increase the number of eggs laid by her hens. The table shows the mean number of eggs laid per hen per year in five generations.

Generation	Mean number of eggs per hen per year
1	180
2	196
3	210
4	222
5	231

(a) Calculate the mean increase per generation in the number of eggs per hen per year, from generation 1 to generation 5. [2]

(b) The increase in egg number became smaller in each generation. Suggest an explanation for this. [2]

(c) Cockerels (male chickens) do not lay eggs. Suggest how Amara could choose which cockerels to use for breeding. [1]

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19 · Part 1

Paper 1 & 2 | Genetic modification (genetic engineering) | [7]

Human growth hormone is a protein. Bacteria can be genetically modified so that they make human growth hormone. The human gene for growth hormone is inserted into a plasmid, and the plasmid is then taken up by a bacterium.

(a) State the type of enzyme used to cut the gene for growth hormone out of human DNA. [1]

(b) The same enzyme is used to cut open the plasmid. Explain why the same enzyme is used. [2]

(c) State the role of ligase in this process. [1]

(d) Give the name for DNA that contains DNA from two different organisms, such as the plasmid with the human gene. [1]

(e) Explain why the bacterium that takes up this plasmid makes human growth hormone. [2]

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20 · Part 1

Paper 1 & 2 | Genetic modification (genetic engineering) | [7]

Many people with diabetes inject insulin. In the past, insulin was extracted from the pancreases of pigs. Today, most insulin is human insulin made by genetically modified bacteria grown in a fermenter.

(a) A fermenter is started with 4.0×10^3 genetically modified bacteria. The bacteria divide once every 30 minutes.

Calculate the number of bacteria in the fermenter after 4 hours. Give your answer in standard form.

Use the formula: number of bacteria = starting number × 2^n, where n is the number of times the bacteria have divided [3]

(b) Explain why genetically modified bacteria grown in a fermenter can produce large amounts of human insulin. [3]

(c) Suggest one advantage of using insulin made by genetically modified bacteria rather than insulin extracted from pigs. [1]

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21 · Part 1

Paper 1 & 2 Genetic modification (genetic engineering) [8]

Caterpillars feed on the leaves of maize plants. Scientists have produced genetically modified (GM) maize containing a gene from a bacterium. The gene makes the maize produce a protein that is toxic to caterpillars.

The table compares GM maize with non-GM maize grown in the same area.

	Non-GM maize	GM maize
Yield / tonnes per hectare	8.0	9.2
Insecticide sprayed / kg per hectare per year	2.5	0.4

(a) Calculate the percentage decrease in the mass of insecticide sprayed when GM maize is grown instead of non-GM maize.

Use the formula: percentage change = (new value − original value) ÷ original value × 100 [2]

(b) Explain why the GM maize has a higher yield than the non-GM maize. [2]

(c) Discuss the advantages and disadvantages of growing this GM maize. [4]

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22 · Part 1

Paper 1 & 2 Genetic modification (genetic engineering) [6]

Weeds are plants that grow among crop plants. Herbicides are chemicals that kill plants.

Scientists have produced genetically modified soya plants that contain a gene from a soil bacterium. The gene makes the soya plants resistant to a herbicide. Farmers can spray the whole field with the herbicide: the weeds die but the soya plants survive.

(a) Explain why killing the weeds increases the yield of the soya crop. [3]

(b) Suggest two possible disadvantages of growing herbicide-resistant soya. [2]

(c) The seeds produced by the genetically modified soya plants grow into plants that are also resistant to the herbicide. Suggest why. [1]

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23 · Part 1

Paper 2 only

Cloning

[8]

Greenleaf Nursery uses micropropagation (tissue culture) to produce large numbers of an orchid with large, brightly coloured flowers.

(a) Describe the process of micropropagation. [4]

(b) Every 5 weeks, each plantlet can be divided to give 6 new plantlets.

Calculate the number of plantlets that can be produced from one plantlet after 20 weeks.

Use the formula: number of plantlets = 6^n, where n is the number of 5-week cycles [2]

(c) All the orchids produced are genetically identical. Explain one disadvantage of this for the nursery. [2]

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24 · Part 1

Paper 2 only

Cloning

[7]

Dolly the sheep was the first mammal to be cloned from a mature body cell.

- The mature cell was taken from the udder of an adult sheep, sheep X.
- The egg cell was taken from a second sheep, sheep Y.
- The embryo was placed in the uterus of a third sheep, sheep Z, the surrogate mother.

(a) Describe the stages in the production of Dolly. [4]

(b) Scientists compared Dolly's DNA with the DNA of sheep X, Y and Z. Dolly's DNA matched the DNA of sheep X only. Explain why. [2]

(c) The nucleus taken from the udder cell is diploid. State what is meant by diploid. [1]

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25 · Part 1 Paper 2 only Cloning [7]

Scientists inserted a human gene into a cell taken from a female goat. The gene codes for a human protein used to treat a blood-clotting disorder. Goats produced from this cell make the human protein in their milk. The scientists then used the cell's nucleus to produce a group of cloned transgenic goats.

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(a) Explain why the scientists produced more of these goats by cloning rather than by breeding a transgenic goat with a normal goat. [2]

(b) Each goat produces 2.0 dm^3 of milk per day. The milk contains 3.0 g of the human protein per dm^3. Each goat is milked on 300 days per year.

The hospital that buys the protein needs 9.0 kg of it each year.

Calculate the number of goats needed to supply the hospital. [3]

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(c) Suggest one advantage of the human protein being produced in the goats' milk. [1]

(d) All of the cloned goats are female. Explain why. [1]